

STRING · Natural-language reasoning audit

Paste prose to inspect claims, evidence, assumptions, and reasoning structure.

Start an audit

Source text

A meta-analysis included seven studies with a total of 604 clinical medical students and compared combined case-based and problem-based learning with lecture-based learning. The combined strategy was associated with significantly higher theoretical examination scores, practical-skills scores, learning satisfaction, and clinical-thinking scores. The estimated effects were large across several outcomes. However, the included studies were conducted in limited educational settings, and the summary does not establish whether outcome assessors were blinded, whether publication bias was excluded, or whether the observed score improvements persisted over time. Examination scores and learner satisfaction may not directly represent improved patient care or diagnostic accuracy. Nevertheless, because the intervention improved every measured outcome, combined case-based and problem-based learning should replace lecture-based teaching across all medical schools.

Claim to audit (optional)

Combined case-based and problem-based learning should replace lecture-based teaching across all medical schools.

Leave blank to let STRING extract the main claims automatically.

Audit focus (optional)

evidence-to-conclusion strength, generalizability, surrogate outcomes, necessity, scope overreach, causal interpretation, and unsupported policy recommendations

Model

gpt-4.1

This tool audits reasoning structure within the supplied text. It does not verify external medical truth or replace clinical judgment.

Generate reasoning audit

Generating reasoning audit...

|                           |                  |                                   |
|---------------------------|------------------|-----------------------------------|
| Reasoning audit           |                  | Run natural-language-e19f38f27950 |
| EXTRACTED CLAIMS          | SUPPORTING SPANS |                                   |
| 3                         | 6                |                                   |
| ACCEPTED CLAIMS           | REJECTED CLAIMS  |                                   |
| 2                         | 1                |                                   |
| GROUNDING STATUS          |                  |                                   |
| COMPOSITIONALLY_SUPPORTED |                  |                                   |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted claims |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Claim 1          | <p>Combined case-based and problem-based learning is associated with higher theoretical examination scores, practical-skills scores, learning satisfaction, and clinical-thinking scores compared to lecture-based learning in the included studies.</p> <p>COMPOSITIONALLY_SUPPORTED    ACCEPTED</p> <p>Supporting spans: source-1, source-2, source-3</p> <p>Why: Sources 1, 2, and 3 together report that the combined strategy was associated with significantly higher scores and satisfaction in the included studies.</p> <p>Refined claim: Combined case-based and problem-based learning was associated with higher examination scores, practical skills, satisfaction, and clinical-thinking scores compared to lecture-based learning in the included studies.</p> |
| Claim 2          | <p>The included studies have limitations in generalizability, outcome assessment, and do not establish whether improvements persist over time or translate to improved patient care or diagnostic accuracy.</p> <p>COMPOSITIONALLY_SUPPORTED    ACCEPTED</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Supporting spans:** source-4, source-5

**Why:** Sources 4 and 5 explicitly state concerns about generalizability, lack of blinding, unknown persistence of effects, and that measured outcomes may not reflect patient care.

**Refined claim:** The evidence comes from studies with limited generalizability and does not establish whether improvements persist over time or translate to improved patient care.

**Claim 3**

Combined case-based and problem-based learning should replace lecture-based teaching across all medical schools.

**OVERSTATED    REJECTED**

**Supporting spans:** source-6

**Why:** Source 6 makes this recommendation, but sources 4 and 5 highlight major limitations that undermine the justification for such a broad policy change.

**Hidden assumption:** Improvements in examination scores and satisfaction are sufficient to justify universal curriculum change.; Findings from limited settings generalize to all medical schools.

Supporting text spans

**source-1**

A meta-analysis included seven studies with a total of 604 clinical medical students and compared combined case-based and problem-based learning with lecture-based learning.

Referenced by: Claim 1

**source-2**

The combined strategy was associated with significantly higher theoretical examination scores, practical-skills scores, learning satisfaction, and clinical-thinking scores.

Referenced by: Claim 1

**source-3**

The estimated effects were large across several outcomes.

Referenced by: Claim 1

**source-4**

However, the included studies were conducted in limited educational settings, and the summary does not establish whether outcome assessors were blinded, whether the intervention was excluded, or whether the observed score improvements persisted over time.

Referenced by: Claim 2

**source-5**

Examination scores and learner satisfaction may not directly represent improved patient care or diagnostic accuracy.

Referenced by: Claim 2

**source-6**

Nevertheless, because the intervention improved every measured outcome, combined case-based and problem-based learning should replace lecture-based teaching across all medical schools.

Referenced by: Claim 3

Assumptions

**claim-3-assumption-1    UNSUPPORTED    from claim-3**

Improvements in examination scores and satisfaction are sufficient to justify universal curriculum change.

**claim-3-assumption-2    UNSUPPORTED    from claim-3**

Findings from limited settings generalize to all medical schools.

Refined claims

**1    COMPOSITIONALLY\_SUPPORTED    from refined-claim-1**

Combined case-based and problem-based learning was associated with higher examination scores, practical skills, satisfaction, and clinical-thinking scores compared to lecture-based learning in the included studies.

**2    COMPOSITIONALLY\_SUPPORTED    from refined-claim-2**

The evidence comes from studies with limited generalizability and does not establish whether improvements persist over time or translate to improved patient care.

Conclusion

**Grounding:** COMPOSITIONALLY\_SUPPORTED    **Acceptance:** ACCEPTED    2 accepted claims used    1 rejected claim excluded    Assumptions remain

Combined case-based and problem-based learning is associated with higher examination scores, practical skills, satisfaction, and clinical-thinking scores compared to lecture-based learning in the included studies, but limitations in generalizability and outcome measures preclude recommending universal replacement of lecture-based teaching.

### Reasoning graph

